

Addressing the Enduring Gender Gap in the Chilean labour Market

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Executive Summary

Despite gender parity in health and education, a stable and growing economy, and indeed a closing gender gap in employment over the past two decades, female labour participation in Chile remains over 20 points below that of men. In the wake of COVID-19 which impacted historically female dominated sectors such as the service industry particularly hard, as well as the socio-political turbulence of the past several years which has led to a renewed interest in feminist and gender-inclusive policy, we ask whether targeted policy efforts can improve conditions and prospects for the female labour market in Chile and ultimately boost economic growth? In this policy note, we will address this gender gap and provide concrete recommendations for Chile's Ministerio del Trabajo y Previsión Social, the Ministry of labour in the Chilean Government, of: 1) subsidizing child care, 2) incentivizing women in STEM education, and 3) using incentives and advertising to change social norms. Through these initiatives, we can both immediately and in the long-term continue to close the gender gap in labour participation as well as labour income.

Keywords: Macroeconomics, labour Market, Employment, Unemployment, Gender, Sex, Chile

1 INTRODUCTION

The Chilean economy is often spoken of with high praise as a modern success story. Indeed since the 1990s, it has experienced sustained economic stability and consistent growth, unlike many of its neighboring countries. Today, Chile is one of only two continental Latin American nations to be considered high-income by the World Bank. Nevertheless, like many of its neighbors in the region, Chile exhibits signifi-

cant socio-economic inequalities. In addition to high levels of inequality in income and wealth, educational opportunities, and health indices, there exist striking gender-based disparities.

Despite having reached nearly complete gender parity in health and survival and educational attainment, Chile ranks only 111th globally in terms of the gender gap in economic participation and opportunity (Global Gender Gap Report, 2020). While the female labour participation rate has increased over 10 percentage points from mid-30s in 1995 to mid-40s in 2020, at 20 percentage points below the male participation rate, it remains below the Organisation for Economic Co-operation and Development (OECD) national average and that of other Latin American countries (ILO, 2020; Contreras & Plaza, 2010).

Stylized Fact 1

The female labour force participation rate in Chile is over 20 percentage points below the male participation rate as well as below the Latin American and OECD average.

Figure 1 highlights how despite far outpacing the rest of Latin America in GDP growth, Chile has not caught up, let alone surpassed, the rest of Latin America in terms of female labour participation. Moreover, of young Chileans neither employed nor studying, an overwhelming 69% of them are women, and of all women who do work, a disproportionate level work in low-income jobs (CASEN, 2017; OECD, 2021).

In this policy note we will suggest targeted policies that aim at improving conditions and prospects for the female labour market in Chile in the short- and long-run. The policy note will be structured as follows: in Section 2 we motivate the opportunity for gender equity focused policy efforts today; in Section 3 we will review theory linking gender inequality, female labour participation, and economic growth; Section 4 details the specific drivers of the gender gap in labour, both generally and specifically in Chile; Section 5 provides our policy recommendations to Ministry of labour that the

government can implement to address the persisting gender gap in labour participation; Section 6 discusses the limitations of our policy note; and finally, Section 7 concludes. We view the policy recommendations of this note not only as an ethical imperative, but also as a fundamental step towards achieving stronger and sustainable future growth.

2 PRESENT MOTIVATION

Since the OECD and World Bank began recording these statistics, the gender gap in employment and labour participation in Chile has always been similar to or even greater than what it is today¹. Nevertheless, today there is a pressing economic need and generational political opportunity to address the persisting gender gap. As displayed in **Figure 1**, female labour participation dropped nearly five points in Chile from 2019 to 2020. Similarly, female unemployment rose to over double digits for the first time since the last global recession in 2009 (ILO, 2020). As the common refrain goes, *never let a good crisis go to waste*.



Figure 1: Comparing Chilean economic growth to female labour participation, relative to the region, over the last 25 years.

2.1 THE EFFECTS OF COVID-19 ON THE CHILEAN LABOUR MARKET

The COVID-19 induced global recession has wrought broad economic consequences, however it has also specifically highlighted and exacerbated existing demographic inequalities in Chile. Not only have reported stress, mental health problems, and even domestic violence incidents against women increased since the pandemic, but the gender gap in the Chilean labour market has also widened. Albanesi & Kim (2021) have found that during the COVID-19 recession employment losses have been greater for women than men, as compared

¹See the falling difference in labour participation from 2010-2020 in Figure 3

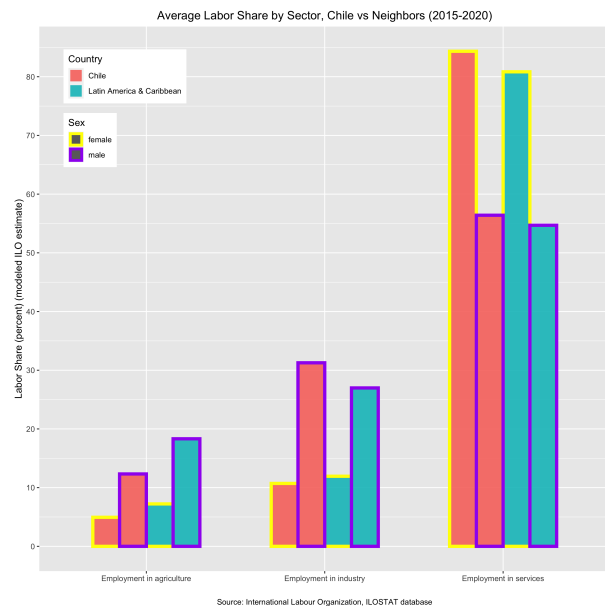


Figure 2: Comparing labour share by gender across aggregate sectors, relative to the region, averaged over the data available since 2015 so as to reflect recent, but not purely COVID-19 impacted figures. Nearly 85% of the female labour force in Chile works in the service sector.

to the opposite trend during the previous three recessions. They argue that the larger employment losses for women can be in part explained by the fact that the service sector – in which women are disproportionately employed and which tends to be less cyclical than other sectors – has been hit particularly hard during this crisis. In **Figure 2** we can see the distribution of employment by gender in aggregate sectors in Chile is consistent with the argument of Albanesi & Kim (2021): of women in the labour force in Chile, nearly 85% work in the service sector, a few points above even the average across all of Latin America and nearly 30 points above the male share. Due to the health crisis and subsequent contagion mitigation measures to contain the spreading of the virus, the service sector faced a dramatic drop in demand. These industries are inherently vulnerable due to high contact intensity, and moreover they are less adaptable to confinement measures as, for the most part, they cannot operate remotely as some other industries can.

Furthermore, Chilean women who became unemployed during the pandemic have largely not re-entered the labour market. While it could yet begin to change, Escobar (2020) found that as many as 80% of all Chilean women who stopped working during the pandemic did not search for re-employment. Indeed, as the OECD report rationalizes, "[i]f the women's unemployment rate increased less than that of men [in the past two

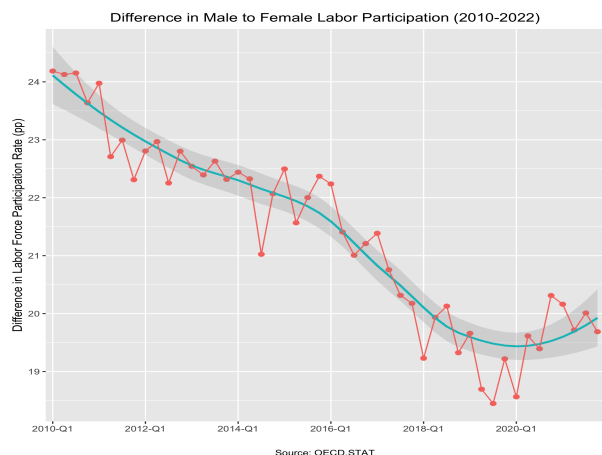


Figure 3: The difference in female-to-male labour force participation had been trending downwards since the 1990s, however has risen since the COVID-19 pandemic.

years], it was because they stopped actively looking for a new job." In other words, while Chilean men have continued to look for work, as reflected in the larger increase in male unemployment relative to the Latin American average, the female unemployment has not risen any faster than the average as many women have left the labour market altogether (OECD, 2021). In **Figure 3** we can see the spike in the labour participation gender gap since the pandemic struck in 2020. This presents a challenge for policymakers, however, as we stress, it also presents an opportunity to build back better and re-engage Chilean women in the labour market.

Stylized Fact 2

Since the COVID-19 induced recession, initial data as per Figure 3 suggest that Chilean men have continued searching for jobs while Chilean women have dropped out of the labour force, reinforcing the *male as the income-earner* dynamic.

2.2 THE POLITICAL MOMENT

In addition to the COVID-19 crisis, the current Chilean socio-political climate presents a further opportunity for gender-equity targeted labour legislation.

Despite being considered one of the most socially conservative countries in Latin America, Chile has a strong feminist movement (Palacios & Martínez, 2006). In recent years, women have led some of the largest demonstrations in Chilean history including protests in 2016, 2017, and 2018 against machismo, domestic violence, and sexual discrimination in the workplace and schools.

Then in 2019, popular uprisings against hyper neoliberal policies and abject lack of economic opportunity for the majority of Chileans culminated in the ongoing re-writing of the Chilean constitution. One of the many significant aspects of this 21st century constitutional reform is that, for the first time in the modern world, the authors of this constitution will be equal parts women and men: in October of 2020, just under 80% of voters in the referendum favored this gender-parity option. Indeed for many critics, the existing constitution written under the dictatorship of Augusto Pinochet remained a barrier to democracy and social equity and served as a lasting symbol of Pinochet-era discouragement of women in the workforce. Now that a new constitution will be written, themes of equality, including gender equity in the labour force, are at the center of discussion.

3 GENDER GAP, OPPORTUNITY COSTS AND ECONOMIC GROWTH

Considerable literature has been published on the relationship between gender inequality, along various dimensions, and economic growth.

Studying gender inequality in empowerment and social norms, Tertilt (2006) showed that transferring of marriage choice from parents to daughters increased per capita output by 60%. Leveraging reforms in the 1970s, Guvenen & Rendall (2015) found that higher divorce risk in the second half the 20th century in the US explains 50% of the rise in female labour supply during this time. Studying gender inequality via exclusion from the labour market, Hsieh et al (2019) found that the fall in misallocation of talent by gender as well as race explains 40% of US GDP growth from 1960-2010. In Europe, Cuberes & Teignier (2016, 2017) found that long-run output losses due to similar gender discrimination are over 15% for OECD countries and slightly greater in developing countries. In India, Esteve-Volart (2009) showed that from 1961-1991 the ratio of female-to-male managers and female-to-male workers were both positively associated with (per capita) output. Also in India, Heath & Tan (2020) showed that from 1983-2005 transfer of income empowered women and increased their numbers in high-paying jobs.

Analyzing panel data for five of Chile's neighbors, Argentina, Brazil, Ecuador, Peru, and Uruguay, Vázquez Rodríguez (2017) finds that a 1-point increase in the female labour force participation rate is associated with a near 6% increase in GDP growth, robust to different specifications and long-term analysis. Indeed, while a comprehensive literature review is outside of the scope of this policy note², it is important to acknowledge the

²For further reading, see Klasen & Lamanna, 2009; OECD,

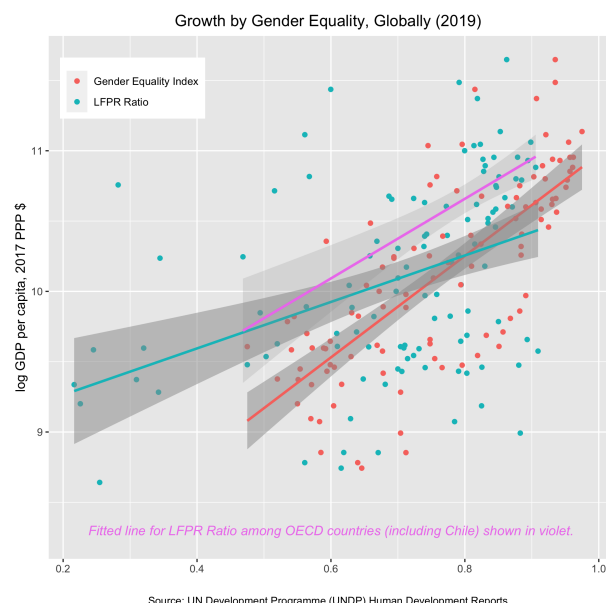


Figure 4: Comparing growth to gender equality across 119 high human development countries around the world we see a strong positive trend, especially with gender equality and LFPR ratio among OECD countries. Here we denote *Gender Equality Index* as the complement of the UNDP Gender Inequality Index (composed of reproductive health, empowerment, and economic status indices) and *LFPR Ratio* as the ratio of female-to-male labour force participation rates.

vast empirical evidence, if partial, confirming the hypothesis that the reduction of a gender gap in labour participation has a positive effect on growth.

Stylized Fact 3

As studied in the literature and shown in Figure 4, there exists a strong positive association between gender equality generally as well as labour participation equality specifically and economic growth.

In addition to empirical evidence, several theories have been postulated to rationalize gender inequality in labour and explain a causal effect between gender inequality and growth. Of particular relevance, we will here consider and later refer back to two models: Galor & Weil (1996) and Cavalcanti & Tavares (2016).

Galor & Weil (1996) model an economy with three factors of production: capital as well as *mental* and *physical* labour. While mental capacity is assumed to be uniformly distributed, men are assumed to have greater physical capacity than women. In addition, as is common, capital and mental labour are assumed

to complement each other more so than capital and physical labour. As a result, *in economies with low levels of capital* women dedicate themselves more to childrearing and domestic work as their opportunity cost on the labour market is lower than men's. As capital-per-worker increases, then, returns to mental labour increase relative to returns to physical labour and women's wages rise relative to men's. Consequently, fertility falls and capital rises, creating a feedback loop that generates accelerated economic growth.

Exogenizing gender discrimination in the labour market, Santos-Silva & Klasen (2021) review the main mechanisms by which this impacts economic growth. Firstly, modelling an economy with partial exclusion of women from the labour force results in women dedicating themselves more to domestic work due to the lower relative opportunity cost of labour, as in Galor & Weil (1996). The lower opportunity cost of children increases fertility which has a well documented negative impact on long-run growth. Secondly, in these models where talent (e.g. mental capacity as in Galor & Weil (1996)) is assumed to be uniformly distributed throughout the population, there is inefficiency due to the misallocation of talent and labour. This reduces innovation, productivity and, ultimately, growth. Thirdly, given partial exclusion from labour, returns to female education are lower. Over time this is endogenized leading to lower human capital investment in girls and ultimately slower development.

Cavalcanti & Tavares (2016) specifically estimate the effect of a gender wage gap on economic growth using a model of the US economy in which women and men are equally productive in the labour market but women receive a fraction of the wage that men receive. They show that the gender wage gap affects growth through two significant transmission channels: **labour supply** and **fertility**³. Firstly, wage discrimination acts as a tax on female labour, reducing female labour and output-per-capita. Secondly, as in Galor & Weil (1996), the lower opportunity cost of labour increases fertility reducing long-run growth. This model can be easily extended by assuming not wage discrimination but discrimination in access to the labour market broadly or access to specific occupations/industries, reaching similar conclusions. As this model is fundamental to our motivation and recommendations, we further elaborate in **Figure 5 and Technical Note 1**.

In short, though *a priori* the strong correlation between gender equality and growth⁴ could be a function of endogenous factors or reverse causality, the extensive

³In fact, the authors find that a 50% increase in the gender wage gap in the US decreases (long-run) per-capita income by 35%.

⁴See 4

literature on the matter suggests the reduction of the gender gap in the labour market indeed has a causal effect on economic growth (Kabeer & Natali, 2013).

Technical Note 1: The effect of gender discrimination in labour on economic growth

We base this note on definitions of a economic growth model based on Solow, and elaborate on the definitions given by Cavalcanti & Tavares. Generally, the production function of the economy with constant returns to scale is given as follows:

$$Y_t = K_t^\alpha (L_t)^{1-\alpha}$$

Where Y denotes output, K denotes capital, A denotes technology and L denotes labour, all in aggregate terms. Capital and labour earn their marginal productivity in terms of rents (in the case of capital) or wages (in the case of labour). Capital accumulation is a factor of economics growth; the more people save, the more the capital stock grows and so the aggregate output grows.

Capital growth is a function of savings s and depreciation δ :

$$K_{t+1} = sF(K_t, L_t) + (1 - \delta)K_t \quad (1)$$

And the labour force, L_t , grows at a rate n . The model in Cavalcanti & Tavares (2016) assumes that women and men have different productivity rates for raising children and supplying labour—women have a comparative advantage in raising children and men in supplying labour on the labour market.

To show this, they introduce a distortionary tax on women, indicating their higher productivity in raising children. This drives a wedge between the productivity and the returns to labour for women, negatively affecting *long-run* output through a decrease in output—as women do not enter the labour market, consumption and ultimately welfare. It also affects output per capita in the short-run, as female labour force participation decreases and consequently, output per capita decreases.

factor in the Chilean society. The gap in labour participation rates of males and females in Chile could hence be linked to the macho culture in this country (Contreras & Plaza, 2010). It seems that even though Chile has relatively high education levels compared to the rest of Latin America, their labour force participation rates do not seem to reflect this higher education level.

Contreras & Plaza (2010) show that generally, a concave pattern is observed in the relationship between age and female labour force participation; younger and older age groups tend to have a lower participation rate compared to middle aged groups. They furthermore find that **women with children below schooling age** also tend to have a relatively low participation rate. When taking into account cultural factors, it seems that women belonging to a machista or conservative environment tend to have lower labour force participation.

Cultural factors in Chile seem to contribute to the gender gap through several factors: firstly, it influences the beliefs of women, in the sense that they feel they do not belong in the paid labour market, but rather should stay home and perform unpaid household tasks; secondly, we can see that women do get paid less when performing paid labour, . These beliefs are structurally ingrained in society, and are hard to change at their core. However, we aim to relieve some of the constraints women face due to these beliefs through two factors: reducing opportunity costs for women through offering childcare programs, and promoting professional fields to women where women are underrepresented.

4.2 ACCESS TO CHILDCARE CENTERS

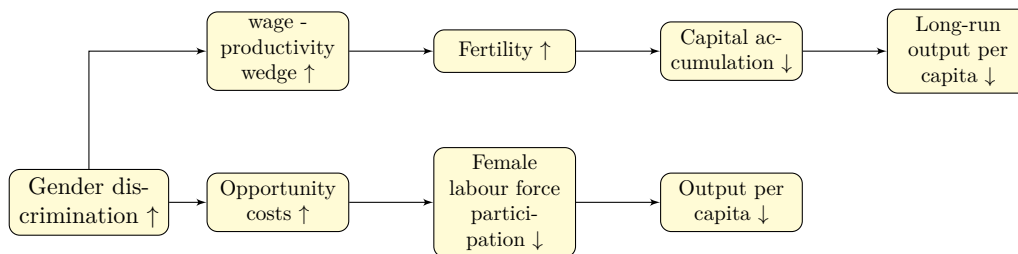
As previously mentioned, it seems that the access of childcare centers play a big role in the labour market decisions for women. In Chile, only 20% of the 0-2 year olds attend daycare—which is below the OECD average of 36% (OECD Family Database, 2021). In comparison, 80% of the 3-5 year olds attends some kind of day care (OECD, 2021), which is significantly higher, however, it is still below the OECD average of 87% (OECD Family Database, 2021). It has to be noted that Chile has made significant improvements in the provision in childcare in the last decade; from 2005 to 2007, there has been a more than doubling in the number of daycare centers in Chile (Contreras, Puentes & Bravo, 2012). Furthermore, several childcare support programs have been put in place. One example is the 4-7 program. A randomized evaluation of this program, which offers three hours of free after-school childcare at public schools for children between the ages of 6 and 13, took place in 2013 and was evaluated by

4 DRIVERS OF THE GENDER GAP IN THE LABOUR MARKET

4.1 CULTURAL FACTORS

In Chile, traditional values are a big part of the culture, also meaning that gender roles are still a promi-

Figure 5: Transition channels of the gender gap on economic growth.



Martínez A. & Perticará (2017). They show that a childcare program offering after-school care increased mothers' employment and participation in the labour force: offering the program increased mothers' labour force participation by 4.3 percentage points, from 60.5 percent in the comparison group, hence a 7 percent increase. The experiment also showed an increase in the use of formal childcare services for children under 6 years old; suggesting that the 4-7 Program effectively lifted the greatest barrier to entering the labour market that women in this subgroup faced—finding childcare for their older children—allowing them to take full advantage of the existing supply of free childcare for younger children. Such a policy can have short term effects. However, this program had no effect on women with less labour market experience, thus additional policies are needed.

Hence, from this we can observe several patterns: even though there has been an increase in daycare centers in Chile, mothers from lower income deciles do not seem to profit from this expansion; secondly, even though there has been a significant expansion of childcare, the enrollment rates of below-schooling age children have been lagging behind, compared to the OECD average; and thirdly, subsidized child care does seem to have a significant effect on the labour participation of women coming from lower income deciles. From this, we observe an opportunity for improvement of the childcare support for lower income mothers, and mothers with children below schooling age. To conclude, we also want to acknowledge that improved provision of childcare would not only improve labour market opportunities for women, but also benefits children in their development (Noboa-Hidalgo & Urzúa, 2012), and in this way could yield long-term effects.

4.3 EDUCATION LEVELS AND FEMALE UNDER REPRESENTATION IN THE SCIENTIFIC SECTOR

Among the drivers of the gender gap in the Chilean labour market are also the differences in education between men and women, caused, among other things, by the cultural factors detailed above (Wang & Degol 2017). Indeed, educational choices, both in terms of

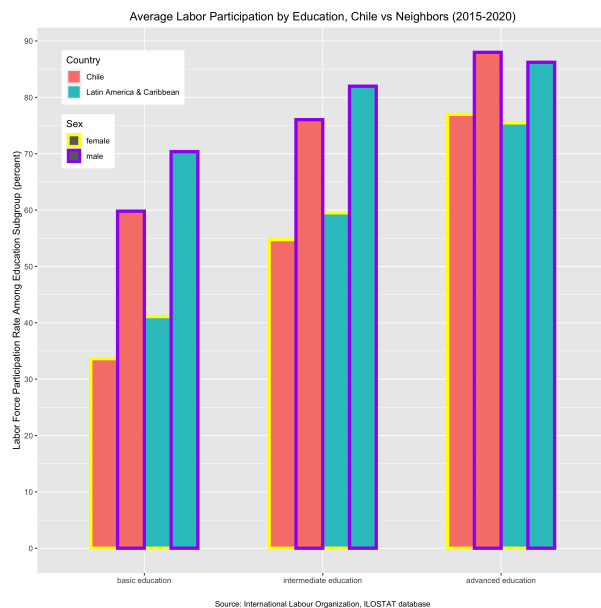


Figure 6: Comparing labour participation rate by gender **across aggregate education levels**, relative to the region, averaged over data available since 2015. Education in Chile is more strongly correlated with labour participation than in the rest of Latin America. *Note that basic education refers to primary or lower-secondary, intermediate to upper-secondary or post-secondary non tertiary, and advanced to tertiary, Bachelor's and above.*

level and field, differ greatly between men and women in Chile: women are still on average slightly less educated than men (although this gap has been narrowing in recent years and is smaller than in other Latin American countries). This gap is problematic for two reasons. First, because lower levels of education are associated with lower levels of labour force participation (**Figure 6**). On the other hand, because lower levels of education are associated with lower levels of pay, thus these differences in educational attainment contribute to the gender pay gap in the Chilean labour market.

However, inequalities in access to education between men and women in Chile are not simply a gap in

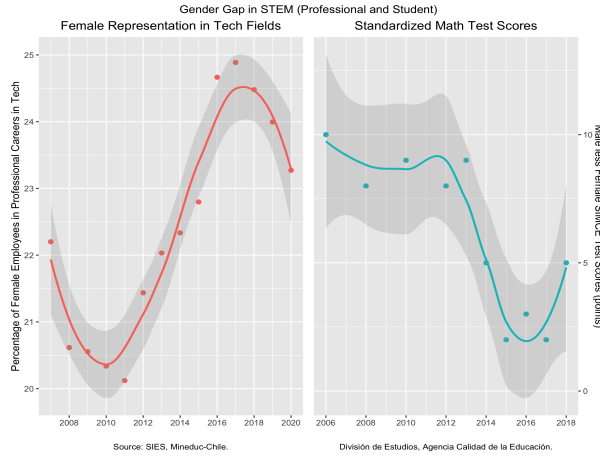


Figure 7: We observe a massive gender differential in STEM employees, with women comprising less than a quarter of professionals in technological fields. Moreover, the differential among professionals as well as the differential in standardized mathematics (SIMCE) test scores for primary and secondary students has been rising in the last several years.

terms of educational attainment. The composition of studies, i.e. the choice of field of study, varies enormously between the genders, and in particular women are underrepresented in science, technology, engineering and mathematics (STEM) studies. In fact, women spontaneously go less into this sector because cultural stereotypes typically portray the STEM sector as male gender-typed. Those stereotypes sometimes result in discrimination against women (in hiring, for promotions, etc.) and generate a social identity threat (i.e. women’s (justifiable) concerns about their group’s devaluation in STEM). This field is also under investigated by women because of the ignorance of what these careers entail, and of a lack of encouragement from parents and teachers. Furthermore, the fact that there are too few women mentors or role models for younger generations maintains this vicious cycle of underrepresentation (Moss-Racusin et al 2020; Roberts 2014). Then, because women do not engage in this type of study, they are also underrepresented in this sector of the labour market (**Figure 7**). As mentioned in Section 3, this causes inefficiency due to the misallocation of talent and labour. Having too few women in science, technology, engineering and math (STEM) fields ultimately reduces innovation, productivity and growth, and further widens the disparity in labour income (as of now, women earn less than 38% of total labour income in Chile (World Inequality Report, 2022)). Thus, better coordination and integration of STEM activities can simultaneously reduce the gender gap and improve economic outcomes (Williams, 2011; Zenobia, 2018; Hsieh et al, 2019).

5 POLICY RECOMMENDATIONS

Our main focus is on increasing the female participation rate on the labour market as well as increasing the labour income share. To do so, we need to focus on policies directed at different groups of females in the labour market. From our previous arguments, we will set up 2 sets of policy recommendations targeting 3 different groups: (1) women with children, and for which this prevents them to enter the labour market; (2) young women that have not yet entered the labour market, and who are at the point of choosing the discipline they want to continue their education. We will also suggest policies such as incentives, and quotas that can be seen as complementary to the main ones we first recommend.

5.1 SUBSIDIZED CHILD CARE

Childcare duties are often cited as one reason why labour force participation is typically lower for women than men. Then, for the first group, we will focus on policies that make services for personal care more accessible, such as providing free after-school care for children and daycare centers for children. For women in the lowest income deciles, child care could be fully subsidized, as they are more likely to take up unpaid care compared to more costly childcare (Anderson & Levine, 1999).

There are numerous papers covering the relevance of childcare in maternal labour participation in several other countries. Lee & Lee (2014) examine the role of childcare availability on the FLFP in Japan. In the case of Japan, FLFP has been increasing, however, there has been a shortage of childcare centers resulting in children being wait-listed for childcare. They find that the availability of child care centers does not affect the fertility choice of Japanese women, however, they do find that a lack of childcare arrangements discourages women, particularly of the age group 30-44, to re-enter the labour market after child birth (Lee & Lee, 2014).

Berlinski & Galiani (2007) study the effects of an extensive expansion of pre-primary school facilities in Argentina taking place between 1994 and 2000. The project constructed 175,000 preschools, an 18% increase compared to the initial number of preschools. They find that enrollment rates of 3-5 year olds increased female labour force participation significantly. Furthermore, they find that the childcare subsidy provided by the program has a positive effect labour supply of mothers of young children, approximately increasing working hours by 2.24 to 4.5 hours per week, based on a 32-hour workweek.

On the **short term**, subsidies for child care should be put in place for all women with children on the labour market. Such a policy could take the form of extending the government-sponsored 4-7 Program to all public schools in the country which do not yet benefit from any other fully subsidized after-school program. Specifically this program is implemented on a small scale – the 4-7 program was implemented in only 87 schools in 2012 (Martínez A. & Perticará, 2017). Hence, there is scope for up-scaling.

In addition to this, we strongly recommend to expand childcare programs to pre-primary school children of ages 0-2 years, in order for Chile to catch up with the OECD average of pre-primary school enrollment rates. This program should be focused on the group who would benefit the most of this policy – mothers of below-schooling age children, coming from low-income deciles, and should offer fully subsidized childcare. Ideally this program would be put in place nationally, and advertised by the national government to raise awareness to the most disadvantaged groups.

Based on Contreras & Plaza (2010) who found that Chilean women with children below schooling age (below 4 years old), were the least likely to enter the labour market, and considering the very low enrollment rates in Chile compared to the OECD average, this would be our *first best* policy implementation.

As an addition to subsidizing childcare to make it less costly for women, it may not be sufficient to ultimately increase the female labour force participation. It may be the case that child care centers are not easily accessible, or located far from the workplace or home. Contreras, Puentes & Bravo (2012) note that despite the significant increase in daycare centers, labour participation did not change, especially in the case of women from lower income deciles. Hence, care centers should be placed in more accessible areas. As we see from several cases (e.g. van Ham & Mulder 2005; Contreras, Puentes & Bravo, 2012), care centers should be located either close to residential areas or close to areas with high economic activity – and will in this way be close to the workplace, to stimulate female labour force participation. Even though there has been empirical evidence that accessibility would positively affect participation, case studies have mainly focused on the cost reduction of increasing child care centers, rather than investigating the impact of improved geographical access.

We propose, based on the available evidence showing that accessibility plays a crucial factor, that programs should be initiated on *improved geographical access to childcare centers* on a local scale. If case studies indeed verify that this has a positive effect on female labour force participation, we propose an expansion of child

care centers, being placed economically active areas, and potentially other areas, depending on the results of the case studies. We would indicate a **long-term** implementation of this program.

Policy intervention 1: Childcare

Reducing childcare costs and improving access to childcare centers, prioritizing low-income mothers of below-schooling age children.

- *Implementation timeline*: short-run in the case of subsidized childcare; medium-run in the case of improving childcare access
- *Impact timeline*: medium-run: women will be incentivised immediately to enter the labour market. Given that programs run for at least several years, and childcare is the major obstacle keeping women from the labour market, women will likely remain active in the labour market for several years.
- *Costs*: medium to high in both the case of fully subsidized childcare and increasing childcare access in areas employing high economic activity.
- *Attainability*: high in the case of subsidies: previous programs have shown positive effects; medium in the case of access to childcare: may be costly and there is a lack of evidence from case studies. However, studies show potential for this to be highly effective.

5.2 SUPPORT AND INCENTIVES IN EDUCATION

As we saw in the previous section, women are not very likely to enter the STEM sector, which is promising in terms of growth and pay. Thus, our second policy recommendation is to encourage and enable women to engage in scientific studies.

To do so, the Chilean government must address the problems of misinformation, under-representation and identity threat detailed in section 4.3.

As this under-representation of women in the STEM sector is not specific to Chile but is observed all over the world, including in developed countries, there is no shortage of international examples, and many of the measures that have been shown to be successful can easily be implemented in Chile.

In order to promote a sustainable presence of women in the STEM sector in the labour market, policies need to be put in place at two levels. We need to develop interest in STEM field among young girls, and we have to assist young women in their STEM studies.

Young girls' interest in the STEM sector can (and should) be stimulated at school. This can be done by

taking into account differences in preferences in terms of teaching style. Indeed, researchers have identified gender differences in the preferred styles of study in mathematics and sciences. In particular, a Finnish study, as part of the Gender Issues, Science Education and Learning (GISEL) project, found that female students preferred engaging in visits to industry and museums, as well as engaging with experts in their science classes, more than the normal science classes they had experienced. Promoting these types of activities can therefore be beneficial to young girls' interest in the field (Roberts 2014).

Thus the Ministry of labour in collaboration with the Ministry of Education should *implement innovative programs* starting in primary and secondary school, including hosting female students in engineering firms, sponsoring visits to RD laboratories, university classrooms and conferences. This can be done on the existing basis of the W-STEM program that took place in Chile, which consisted of the establishment of 49 training centers to train women to intervene in schools and promote scientific careers to young girls.

Several programs have already successfully implemented such measures. Google's Mind the Gap! program in Israel succeeded in motivating 40% of the girls who participated to choose computer science as a major stream in upper secondary school. A similar program in Japan has also had positive effects, with one prefecture reporting an increase of 50 percent in the numbers of school girls choosing STEM courses in higher education between 2002 and 2011. Such programs also exist in France, Germany, Australia, and elsewhere, with similar results, though often difficult to measure accurately.

In addition to this reform of science education, the Chilean education system should work to *improve its representation of women in the STEM sector*. As even brief exposure can spark female students' interest in STEM (Stout et al., 2011), initiatives such as putting pictures and information about female scientists into STEM classroom spaces and lectures can help young girls develop an interest and project themselves in this field. In addition, high-quality videos featuring successful female STEM role models could be shown to students during class lectures, at larger events, or online. Those new and creative techniques can make role models broadly accessible at small cost and without creating additional burdens on female scientists (Moss-Racusin et al 2020).

In addition to motivating girls to study in STEM, there is also a need to *facilitate the academic careers of female students in this sector*, and to counteract the weight of stereotypes they face. This can be done with financial incentives (Roberts 2014).

Universities in Chile should be provided with greater national governments funds for the appointment of female professors. Projects should receive higher evaluations during funding selection processes according to the number of female researchers involved (for instance 5% higher evaluation for each woman involved). Measures, such as scholarships or fellowships specifically reserved for female students and researchers should be put in place.

Such programs have been positively correlated with the proportion of women in professorial roles in Switzerland and Greece.

Policy intervention 2: Education

Developing the interest of young girls for the STEM sector and financially helping young women pursuing their studies in this field.

- *Implementation timeline:* short run (potentially immediate)
- *Impact timeline:* medium long run (engaging today's girls in the STEM sector will only have an impact on the labour market in 5-15 years)
- *Costs:* low to medium for both groups of women
- *Attainability:* medium to high (there is an inherent uncertainty to tasks such as changing social norms and shaping mentalities, but the large body of evidence suggest high rates of success)

5.3 RAISING AWARENESS OF PROGRAMS

Complementary to these policy recommendations, we do recognise the importance of raising awareness of the existence of these support programs. We have emphasized the role of promoting female representation in the STEM sector. However, we do want to add that, since many of these policy recommendations are aimed at relatively disadvantaged groups who may not be aware of these opportunities, campaigns and advertisements for these programs (and existing programs) should be put in place as a complement to these recommendations.

6 LIMITS OF OUR POLICY NOTES

The possible limitations of our policy note lie in the fact that 1) we do not specifically address the problem of discrimination in the Chilean labour market; 2) we do not make recommendations for women lacking skills; 3) we leave aside the policy of quotas, which is relatively classic in programs to combat gender inequality.

We have decided to focus on the problem of human capital disequilibrium between men and women, and to leave aside the possibility that men and women may have the same human capital but that their access to or their experience on the labour market differ. We did so as directly assessing and addressing discriminatory practices requires very specific and reliable data that could not be found for Chile.

We also did not address the issue of women on the labour market who are lacking skills. This mismatch problem could be addressed by policies such as job-training. We renounced properly formulating such a policy as doing so requires a very specific knowledge of the structure and the needs of the labour market which could not be gathered for the Chilean labour market. Moreover, it would be difficult to design something of the sort only for women as men can also suffer from inadequate training. A job-training policy recommendation could however be a future extension to our policy note.

We also decided to leave aside quotas. We did so as this policy instrument has mixed (if not negative) results in the literature: Roberts (2014) underlines the downsides of quotas as they induce a (perceived) discrimination against men, neglect the cultural elements of the issue, and fail to organically and culturally instil the practice of equality, furthermore quotas fail to influence the private sector which is in general the most unequal; Pande et al. (2011) found that corporate board quotas was adversely influencing short-run profits.

7 CONCLUSION

This policy note proposes policy recommendations aimed at reducing or eliminating the gender gap in the Chilean labour market. Although the gender gap in Chile has narrowed in recent decades, women's participation in the labour market remains 20 percentage points lower than men's and their earnings 38% lower. In addition, the Covid crisis showed how fragile the gains of recent years were, since after the crisis nearly 80 percent of Chilean women who had lost their jobs did not seek new ones. Gender inequalities in the Chilean labour market are also structural, with far fewer women in the higher-paying science (STEM) sectors. Women make up only 25 percent of workers in this sector and this figure has been stagnant for several decades. Gender inequality in the labour market is not only ethically reprehensible but also detrimental to growth - the country is depriving itself of a large part of its labour force and talent - and it is crucial that the Chilean government address this problem. It is with these figures in mind and in the context of the aftermath of the Covid crisis and the recent political renewal, that we propose two main recommendations. The first, which should be effective in the short term,

is to offer more childcare options, where our first-best option would be a fully subsidized childcare program offered firstly to low-income mothers, then expanding to all mothers of children being of below-schooling age. As an addition to this, general access to childcare in residential and highly active economic areas should be expanded, but this could be pursued in the longer-term. The second, which should be effective in the medium to long term, is to address the under representation of women in the STEM sector, by increasing the interest of young girls in these disciplines and thus encouraging them to pursue studies in this sector, and by ensuring support and financial aid for young women pursuing studies in these disciplines. We hope that these suggestions will provide a key source of inspiration for the Chilean government as they seek to tackle the labour market gender gap and ensure that Chile is able to affirm a status not just as a regional economic model to follow, but as global example of macroeconomic policy success.

Economic terms

- *Opportunity costs*: the foregoing of potential gains of the alternative that is not chosen. In this context, the opportunity cost of women wanting to enter the labour market is for example, caring for their children and taking care of household chores.
- *Labour force participation*: number of people aged 16 and older who are employed or actively seeking employment, divided by the total working-age population.
- *Unemployment rate*: percentage of the total labour force that is unemployed but actively seeking employment and willing to work.
- *Capital accumulation*: the growth in wealth through investments or profits.

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8 APPENDIX

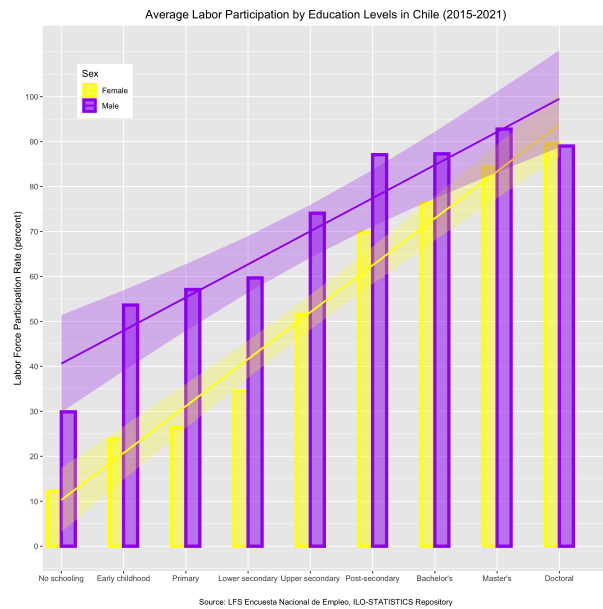


Figure 8: Comparing labour participation rate by gender **across detailed educational attainment levels** averaged over data available since 2015. Not only does female labour participation rise with education, but it also closes the gap to male labour participation.